



Feel like experimenting?

You can get a bunch of lab rats at <http://tinyurl.com/otozvq>

Sleeptalking computers

Scientists at UC San Diego are developing a computer that can partially operate without consuming any electricity

Plasma transistor

A group of mad scientists out to solve the energy crisis have fabricated a transistor out of ionized gas and electrons

How stuff works

sense that one is billions of billions of times weaker than the other. While Einstein puzzled over this, things got a bit worse...

Playing dice

As scientists delved deeper into the atom, they discovered two things: one, that the atom was made up of protons, neutrons and electrons and that the universe has two more major forces — the strong nuclear force, which holds protons and neutrons together in the atom, and the weak nuclear force, which makes atoms radioactive. Secondly, they discovered that at the atomic level, the universe changes completely.

At the subatomic — or quantum-level, the fabric of space-time isn't really a fabric at all. It's bumpy and frothy and it's called quantum foam. It contradicts everything Einstein had believed in — the universe as structured and rational and orderly.

In this quantum foam, there's no such thing as certainty — all events only have a probability. So if you're a little electron and you want to be in Mumbai, there's only a probability that you'll end up there — just like there's a probability you'll end up in Delhi, or Bangalore, or at the top of Everest. In fact, you'll

really end up everywhere, just in different universes — so you could be in Mumbai in this universe, but you're in Delhi in another. These crazy theories — among others form quantum mechanics. For all its bizarreness, quantum mechanics is surprisingly accurate. It explains the behaviour of the universe at the subatomic level, and any experiment that set out to verify any of its predictions has ended in success.

Unfortunately, quantum mechanics and general relativity were horribly incompatible. Quantum mechanics makes perfect sense at the subatomic scale, but doesn't apply to anything bigger. General Relativity, on the other hand, explains all the big stuff, but falls apart at the atomic scale. For the most part, though, this incompatibility didn't seem to cause a major impasse.

But then came the proposition of the black hole — the entire mass of a star, concentrated into a tiny, subatomic point; its gravitational field so strong that even light can't escape it. How do you treat such a formation? Do you apply the laws of quantum mechanics to it because it's so small, or do you use General Relativity because it's so heavy? The same

STRINGS IN THE QUANTUM WORLD

At the subatomic scale, space-time is foamy — so much so, that it's possible that in all that frothy chaos, the fabric of space-time may tear and repair itself on a regular basis.

Einstein's general relativity forbids this; it makes room for the possibility of tears (called wormholes) that exist as a feature of the universe itself — just like your t-shirt has three holes when you buy it — but space-time isn't supposed to rip at will. If it does rip, it may well be the end of the universe.

According to string theorists, however, these rips could exist, and it's the strings that are keeping all hell from breaking loose. As these strings travel through the universe, they're tracing out tubes, which form "bubbles" around these tears in space-time, effectively preventing them from doing any damage.

If space-time can rip at will, we might eventually be able to use these rips to teleport to and from wherever we want. It's a tiny possibility, but it's there, and that's cool enough.

problem applies to the Big Bang — general relativity can talk about what happened just after the Big Bang, but there's no way to talk about the instant of the Bang, when the entire universe was just a single point. So far, any attempt to integrate quantum mechanics with general relativity has resulted in nonsense.

But if quantum mechanics applies everywhere, and General Relativity applies everywhere, why don't they play well together? There had to be some way to unify the universe's forces...

Little giants

As scientists started smashing atoms together, they discovered that matter didn't end at protons and neutrons — these subatomic particles were made of even smaller particles, called quarks. Naturally, the scientists kept smashing, and unearthed a whole mess of particles — enough to develop the Standard Model of particle

other. Similarly, protons and neutrons exchange gluons to stick together, and bosons are the reason that atoms are radioactive. This model of the universe ties electromagnetism, the strong nuclear force, and the weak nuclear force together with a neat little bow. Even better, there's plenty of experimental evidence to suggest that this is, indeed, how the universe works.

But what of gravity? While the Standard Model explains so much, it still falls short of providing a plausible explanation for gravity.

That's where string theory comes in.

The crazy yarn

String Theory, as we mentioned before, says that even the most fundamental particle is made up of billions of little strings that can stretch, compress, and vibrate. Just like different vibrating guitar strings produce different sounds, the way these strings vibrate determines the nature of the particle they form.

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physics.

According to the Standard Model, the forces of the universe are all caused by an exchange of fundamental particles, called messenger particles — magnets are attracted to iron, for examples, because both are throwing photons at each

Most theoretical physicists took an instant dislike to this new theory. There wasn't — and isn't now — any way to prove these strings actually exist — in the world of science, that means that string theory is still more philosophy than science. Worse, it made some really



Inside the bubble chamber of an atom smasher, the fundamental particles of the universe make their existence known. Atoms break into smaller particles, which then follow the different paths you see here